

Bitwise linear trail through **QALQAN** found by SAT

CiVerLy

August 21, 2026

1 QALQAN

Maximal linear correlation: 2^{-28}

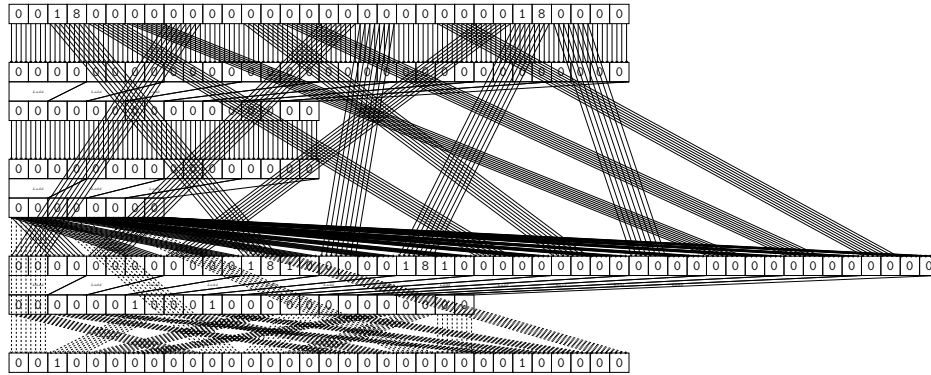
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0	0
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0	0
<i>RspAdd_start</i>																									
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0	0
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0	0
<i>SBox_Layer</i>																									
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0
<i>L</i>																									
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
<i>Add128</i>																									
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
<i>SBox_Layer</i>																									
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
<i>L</i>																									
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
<i>Add128</i>																									
0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0
<i>SBox_Layer</i>																									
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0
<i>L</i>																									
0	0	1	9	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0
0	0	1	9	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0

2 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

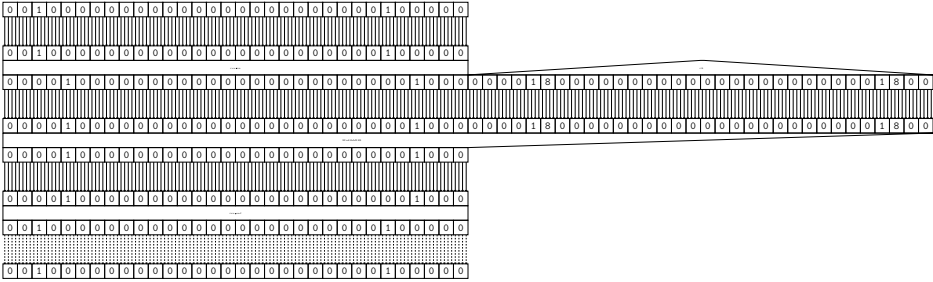
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0
0	0	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	0	0	0
$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{9}{9 \text{ Base}}$	$\frac{4}{4 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{8}{8 \text{ Base}}$	$\frac{7}{7 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	$\frac{0}{0 \text{ Base}}$	
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0

3 L

Maximal linear correlation: $2^{-2.0}$ 

4 Add128

Maximal linear correlation: $2^{-2.0}$

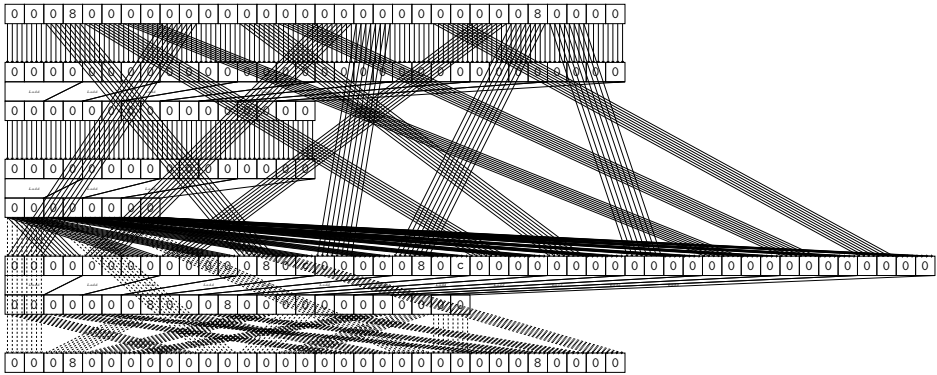


5 SBoxLayer

Maximal linear correlation: $2^{-6.0}$ [illegible]

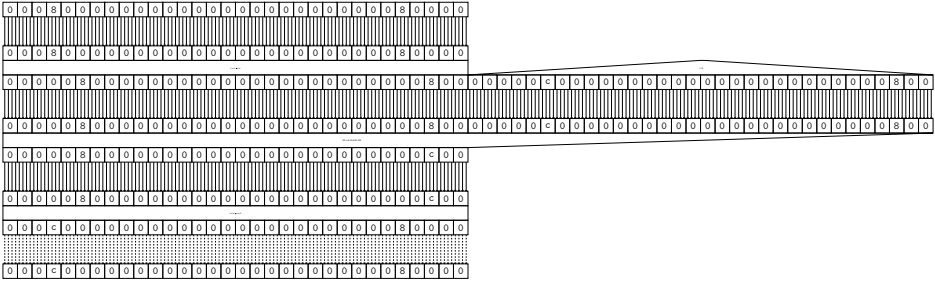
6 L

Maximal linear correlation: $2^{-2.0}$



7 Add128

Maximal linear correlation: $2^{-2.0}$



8 SBoxLayer

Maximal linear correlation: $2^{-6.0}$ [illegible]

9 L

Maximal linear correlation: $2^{-2.0}$

